# Shrutheesh Raman Iyer

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## EDUCATION

**University of California, San Diego (UCSD)** Sep 2021 – Jun 2023
M.S. in Computer Science & Engineering | GPA: 4.0/4.0
*Relevant Coursework: Intro & Math to Robotics, Sensing and Estimation, Computer Vision, 3D Geometry, Convex Opt.*

**R.V. College of Engineering, Bangalore (RVCE)** Aug 2016 – Aug 2020
Bachelor of Engineering in Computer Science & Engineering | CGPA : 9.6/10.0

## SKILLS

- *General Programming*: Python, C/C++, R, MATLAB
- *Software/Tools*: ROS, PyTorch, Moveit, CoppeliaSim, Bullet, PCL, OpenCV, Git, Tensorflow
- *Robotics Hardware*: experience with Fetch Robot, Raspberry Pi, Arduino, Pixhawk, Yaskawa Motomini

## RESEARCH

### Cognitive Robotics Lab, Contextual Robotics Institute, UCSD

**Generalized Framework for Robot Tool Improvisation** | Advisor: Henrik I. Christensen Oct 2021 – Jun 2023

- Master's thesis project work on a planning framework for robots to reason about tool usage using affordances
- Implementing atomic actions (e.g. grasping, pushing) on the Fetch Robot using C++, Python, ROS and MoveIt
- Thesis available at Proquest

**Task and Motion Planning for Home Robot** | Advisor: Henrik I. Christensen Jan 2023 – May 2024

- Part of research work to develop experience-based Task and Motion planning in foliated manifolds, presented at RSS 2024
- Also contributed to the service home robot project in the lab in terms of planning and integration of relevant components
- Relevant Publications : IRC 2023, RSS 2024

### Robert Bosch Centre for Cyber-Physical Systems, Indian Institute of Science *Intern & Technical Associate*

**Teleoperation of Humanoid Robot** | Advisor : Bharadwaj Amrutur, Raghu Krishnapuram Apr 2020 – Jan 2021

(Collaboration with Hanson Robotics and Tata Consultancy Services R&D)

- Research project to develop a teleoperated humanoid robotic nurse as part of ANA Xprize Avatar challenge which aims to deploy real-time senses in remote environments. https://aham-avatar.org/
- Worked on tracking human arm movements on the robot using optical trackers and VR technology. Developed an in-house data glove with finger tracking and haptic feedback for hand teleoperation

## References


[1] J. Hu, S. R. Iyer, J. Wong, and H. I. Christensen. Motion planning in foliated manifolds using repetition roadmap. In *Robotics Science and Systems (RSS)*, Delft, Netherlands, July 2024.

[2] S. R. Iyer, A. Pal, J. Hu, A. Adeleye, A. Aggarwal, and H. I. Christensen. Household navigation and manipulation for everyday object rearrangement tasks. In *International Conference on Robotic Computing (IRC)*, Laguna Hills, CA. IEEE, Dec. 2023.

[3] S. R. Iyer. *Improvisational Robot Tool Use using Affordance based Planning*. Master's thesis, UC San Diego, 2023. Available at `https://escholarship.org/uc/item/8622p2rs`.

[4] S. R. Iyer, V. P. Kuruvilla, R. Krishnapuram, and P. Bhamidipati. Deep-feature-based visual odometry for autonomous emergency parking. In *Proceedings of the 2023 6th International Conference on Advances in Robotics*, pages 1–6, July 2023.

[5] P. R. Iyer, S. R. Iyer, R. Ramesh, M. Anala, and K. Subramanya. Adaptive real time traffic prediction using deep neural networks. *IAES International Journal of Artificial Intelligence*, 2019.

## Experience

### Aurora Inc. — *Perception Software Engineer (Intern Previously)*

**Long range Lidar Detection** — Jul 2023 – Present | Jun 2022 – Sep 2022

- Working on long range lidar object detection for the self-driving autonomy stack
- Responsible for maintaining quality of datasets, improving detector capabilities, developing metrics, and visualization.
- Skills: 3D Object Detection, Lidar Processing, Depth Processing, PyTorch, Python, C++, CUDA

### LightMetrics Pvt. Ltd. — *Research Intern*

**Neural Network Pruning & Automatic Report Generation** — May 2019 – Aug 2019 | Jun 2018 – Jul 2018

- Compare the performance of various techniques for compressing neural networks for real-time performance on edge compute by implementing filter-based and activation-based pruning algorithms
- Evaluated their performance in terms of the speed, accuracy, and size of the neural networks. Achieved 6× speed-up and 12× size reduction with no significant decrease in accuracy for object detection

## Projects

### 3D Reconstruction using SLAM — *Mar 2022 – June 2022*

- In a team of 3, developed an end-to-end VR application for 3D reconstruction and VR rendering of archaeological sites with an inertial RGBD camera using SLAM. https://github.com/KolinGuo/maya-slam
- Primarily contributed to the development and interface of SLAM to obtain 6D pose trajectory
- Implemented using C++, Python, ROS, Unity and Docker; as part of Embedded Systems course CSE237D

### Extended Kalman Filter and Particle Filter for SLAM — *Jan 2022 – Mar 2022*

- Implememted Visual Intertial Simultaneous Localization and Mapping (SLAM) using Extended Kalman Filter (EKF), and Particle Filter (2 independent projects)
- EKF implemented using stereo camera observation model, and IMU measurement model to localize the autonomous robot
- Using Particle Filter, built a 2D occupancy grip map using Laser correlation observation model, and differential drive odometry. Performed texture mapping using RGB information

### Project Jatayu, RVCE — *Computer Vision Head and Team Lead*

**Unmanned Aerial Vehicles with Project Jatayu** — Aug 2017 – Aug 2020

- Created drones with Project Jatayu, the flagship autonomous unmanned aerial vehicle (UAV) team of the RVCE
- Enabled perception in noisy and poorly-lit environments by developing image and video processing protocols for drone footage, and designed communication pipelines.
- Led the team in the AUVSI Student Unmanned Aerial Systems Competition 2019 (SUAS 2019), Maryland, USA

## SOCIETIES & COMMITTEES

### Robograds, UCSD — Jan 2022 – PRESENT

Academic Chair

- Member and later academic chair of Robograds, a graduate student organization for robotics students at UCSD
- Hosted high-school visits, demos, social events, and biweekly talks on recent development in robotics by students and industry

### Project Jatayu, RVCE — Aug 2017 – Aug 2020

Team Lead and Head of Computer Vision Subsystem

- Successfully planned and constructed custom unmanned aerial vehicles for autonomous tasks
- Successfully led a team of over 30 members at the AUVSI Student Unmanned Aerial Systems Competition 2019(SUAS 2019), held in Maryland, USA where we secured 27$^{th}$ rank among 75 participating teams

### RV QuizCorp, RVCE — Aug 2016 – Aug 2020

Coordinator

- Host and organizer of Under The Peepal Tree (UTPT) in 2018, 2019 & 2020, one of Asia's largest quiz festivals
- Organized and participated in inter and intra-college quizzes with 100+ participants